

Serializability & Concurrency Control — DBMS

Definition: Serializability is the correctness criterion for concurrent transaction execution. A non-serial schedule is **serializable** if its execution outcome is equivalent to executing the transactions sequentially one after another (a Serial Schedule).

1. Detailed Technical Explanation

1. Types of Schedules:

- **Serial Schedule:** Transactions execute strictly one after another (No interleaving). Guaranteed to be correct, but offers poor CPU utilization.
- **Concurrent (Interleaved) Schedule:** Operations of multiple transactions interleave. High performance, but requires concurrency control.

2. Conflict Serializability:

Two operations O_i and O_j in a schedule are in **conflict** if and only if:

1. They belong to **different transactions**.
2. They access the **same data item**.
3. At least one of the operations is a **WRITE** ($W(X)$).

Conflicting Operations Matrix:

Operation T_1	Operation T_2	Conflict Status
Read(X)	Read(X)	No Conflict
Read(X)	Write(X)	CONFLICT
Write(X)	Read(X)	CONFLICT
Write(X)	Write(X)	CONFLICT

3. Precedence Graph (Serialization Graph) Test:

A directed graph $G = (V, E)$ used to test Conflict Serializability:

- **Vertices (V):** All transactions in schedule.
- **Edges (E):** Directed edge $T_i \rightarrow T_j$ if T_i performs a conflicting operation on data item X before T_j .

PRECEDENCE GRAPH RULE:

- If Precedence Graph contains NO CYCLES $\rightarrow$ Schedule is CONFLICT SERIALIZABLE.
- If Precedence Graph contains CYCLES $\rightarrow$ Schedule is NOT Conflict Serializable (Concurrency Anomaly!).

Example (Cycle detected):

```
[ T1 ] ---- Read(X), Write(X) ----> [ T2 ]
  ^                                   |
  |----- Write(Y), Read(Y) -----| (Cycle! Not Serializable)
```

2. Core Concepts & Memory Keywords

- **Conflict Equivalent:** Two schedules can be transformed into one another by swapping non-conflicting operations.
 - **Serialization Graph:** Directed graph used to test conflict serializability for cycles.
 - **View Serializability:** Broader class of serializability accounting for blind writes.
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3. Must-Write Points for Exams

- A schedule is conflict serializable if it is conflict equivalent to some serial schedule.
 - Checking for cycles in a precedence graph using Topological Sorting validates conflict serializability.
 - Conflict serializability is a sufficient condition for view serializability.
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4. Quick Recall Flow

Concurrent Operations -> Identify Conflicts (R-W, W-R, W-W) -> Draw Precedence Graph -> No Cycles? -> Serializable!